Roll number: ME23BTECH11036

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MA1220 - Calculus II

Date: October 14, 2023 Maximum Marks 50

Time: 120 minutes, 09:30 AM - 11:30 AM

Q.0. Make an index on page no 1. (On page no.1, write Q.nos and page numbers in two columns). For making an index, you get **(+2 marks)**, and for not making an index, you get **(−2 marks)**. (2 marks)

Q.1 Answer the following. (No justification required). **(Note that you must answer Q.1 on page number 2 only.)** (8 marks)

(a) If $f(x) \geq g(x)$ on the interval $[a, \infty)$ then, if $\int_a^\infty f(x)dx$ converges then $\int_a^\infty g(x)dx$ also converges. **(True/False)**?

(b) $\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial^2 f}{\partial y \partial x}$ is always true. **(True/False)**?

(c) If $f_x = f_y = 0$ at a point (x, y) then f is differentiable at (x, y). **(True/False)**?

(d) $D_{\mathbf{v}}\left(\dfrac{\partial f}{\partial x_2}\right)(-3, 4) = \dfrac{\partial^2 f}{\partial x_1 \partial x_2}(-3, 4)$ if $\mathbf{v} = \langle 1, 0\rangle$ **(True/False)**?

(e) If $f(1) = 12$, f' is continuous, and $\int_1^4 f'(x)\, dx = 17$, then $f(4) = ?$.

(f) Is $\int_{-1}^{1} \frac{1}{x}\, dx$ convergent or divergent?

(g) Describe the domain $\mathcal{D}$ of the function $f(x, y) = \sqrt{1 - x^2 - y^2} \cdot \ln\left(x^2 + y^2 - 1\right)$.

(h) Is it true that $\int_{-\infty}^{\infty} x\, dx = \lim_{t\to\infty} \int_{-t}^{t} x\, dx$. **(True/False)**?

Q.2 Answer the following:

(a) Find the directional derivative of the function $f(u, v) = u^2 e^{-v}$, at the point $(3, 0)$ in the direction of the vector $3\mathbf{i} + 4\mathbf{j}$. (3 marks)

(b) By comparing (without integrating,) determine whether the integral $\int_1^\infty \dfrac{1}{\sqrt{x^3 + 1}}\, dx$ converges or diverges. (2 marks)

(c) Find the local minima, maxima, saddle points of the function $f(x, y) = x^2 + xy + y^2 + y$ in $\mathbb{R}^2$. (3 marks)

Q.3 Answer the following:

(a) Find the volume V obtained by rotating the region enclosed by the graphs of the functions $f(x) = x(5 - x)$ and $g(x) = 8 - x(5 - x)$ about the y-axis. (4 marks)

(b) Find the area of the region bounded by the graphs of $y = \dfrac{8}{x^2}$, $y = 8x$, and $y = x$. (4 marks)

Q.4 Answer the following:

(a) If the region $\mathcal{R} = \{(x, y) \mid x \geq 1,\ 0 \leq y \leq \frac{1}{x}\}$ is rotated about the x-axis, by using improper integral show that:

(i) The volume of the resulting solid is finite, and find the volume. (2 marks)

(ii) The surface area is infinite. (3 marks)

(b) If $f(x, y) = \dfrac{x^{2023}\, y}{x^{4046} + y^2}$, does (3 marks)

$$\lim_{(x,y)\to(0,0)} f(x, y) \text{ exist?}$$

Q.5 Answer the following:

(a) Determine the function $f(x)$ and all values of a such that (2 marks)

$$\int_a^x f(t)\, dt = x^2 + x - 6.$$

(b) Evaluate $g''(\frac{\pi}{6})$, where (2 marks)

$$g(y) = \int_3^y f(x)\, dx \quad \text{and} \quad f(x) = \int_0^{\sin x} \sqrt{1 + t^2}\, dt.$$

(c) Using Riemann integration, find the value of (2 marks)

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n}{n^2 + k^2}.$$

(d) Does there exists a function $f(x, y)$ such that $f_x(x, y) = xy$ and $f_y(x, y) = x^2$? If yes, find f. If no, explain your reason. (2 marks)

Q.6 Answer the following:

(a) Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be a function defined by (4 marks)

$$f(x, y) = \sigma x + \pi y \quad \text{for all } (x, y) \in \mathbb{R}^2$$

where σ and π are constants. Prove (**using $\epsilon - \delta$ definition**) that f is continuous on $\mathbb{R}^2$.

(b) Let $g : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be a function defined by $g(x, y) = 5x + 4y^2$. Prove (**using the steps given below**) that $g(x, y) = 5x + 4y^2$ is differentiable at $(2, 1)$. (4 marks)

(Step 1.) Compute linearization function $L(x, y)$ (the equation of tangent plane) of $g(x, y)$ at $(2, 1)$.

(Step 2.) Compute the difference $g(x, y) - L(x, y)$.

(Step 3.) Show that for all $(x, y) \neq (2, 1)$, one can write

$$0 \leq \frac{|g(x, y) - L(x, y)|}{\sqrt{(x-2)^2 + (y-1)^2}} \leq A|y - 1|$$

for some constant A. Find one such A.

(Step 4.) Apply the definition of differentiability for $g(x, y)$ at $(2, 1)$.